Notation (Consistent)

λ, μ, α	rates (events/time)
$X(t)$	state of CTMC at time t
$N(t)$	# events by time t (counting proc.)
X_i	i-th interarrival time
$S_n = \sum_{i=1}^n X_i$	n-th arrival/event time
v_i	total rate out of state i in CTMC
q_{ij}	transition rate $i \to j$ $(= v_i P_{ij})$
$P_{ij}(t)$	$\mathbb{P}(X(t+s) = j \mid X(s) = i)$
P_{ij}	embedded chain trans. prob.
π_j, P_j	limiting/stationary probability
Q	generator matrix
ρ	traffic intensity $\lambda/(s\mu)$
L, L_q	avg # in system / in queue
W, W_q	avg time in system / in queue
F_e	equilibrium distribution
$\gamma(t), \delta(t)$	excess life, age at time t

Exponential Distribution

$X \sim \text{Exp}(\lambda)$: rate λ, mean $1/\lambda$.

$f(x) = \lambda e^{-\lambda x}, \quad F(x) = 1 - e^{-\lambda x}, \quad \bar{F}(x) = e^{-\lambda x}$

$\mathbb{E}[X] = 1/\lambda, \quad \text{Var}(X) = 1/\lambda^2, \quad \mathbb{E}[X^n] = n!/\lambda^n$

Memoryless Property

$$\boxed{\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t) = e^{-\lambda t}}$$

Only continuous dist. with this property. Means: "remaining life doesn't depend on elapsed time."

Competing Exponentials

$X_i \sim \text{Exp}(\lambda_i)$ indep., $i = 1, \ldots, n$. Then:

$$\boxed{\min(X_1, \ldots, X_n) \sim \text{Exp}(\lambda_1 + \cdots + \lambda_n)}$$

$$\boxed{\mathbb{P}(X_j = \min) = \frac{\lambda_j}{\sum \lambda_i}}$$

Identity of min is **independent** of value of min.

After X_j occurs first, survivors still have original Exp distributions (memoryless).

Poisson Process

Definition (3 equiv. forms)

Def 1: $N(0) = 0$; indep. increments; $N(t) - N(s) \sim \text{Poi}(\lambda(t-s))$.

Def 2 (infinitesimal): $N(0) = 0$; indep. incr.; $\mathbb{P}(N(h) = 1) = \lambda h + o(h)$; $\mathbb{P}(N(h) \geq 2) = o(h)$.

Def 3: Interarrivals $X_1, X_2, \ldots \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$.

Key Properties

$N(t) \sim \text{Poi}(\lambda t)$: $\mathbb{P}(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}$

$\mathbb{E}[N(t)] = \text{Var}(N(t)) = \lambda t$

Waiting time: $S_n \sim \text{Gamma}(n, \lambda)$, $f_{S_n}(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}$

$\mathbb{E}[S_n] = n/\lambda, \quad \text{Var}(S_n) = n/\lambda^2$

Order Statistics Property

Given $N(t) = n$, the n arrival times $(S_1, \ldots, S_n) \sim$ order statistics of n iid $\text{Unif}(0, t)$.

$\mathbb{E}[S_k \mid N(t) = n] = \frac{kt}{n+1}$

$\mathbb{E}\left[\sum_{i=1}^{N(t)} S_i\right] = \frac{\lambda t^2}{2}, \quad \mathbb{E}\left[\sum_{i=1}^{N(t)} f(S_i)\right] = \lambda \int_0^t f(s)\, ds$

Proof idea: $\mathbb{P}(S_1 \leq s \mid N(t) = 1) = s/t$ (uniform).

Superposition (Merging)

Indep. $\text{Poi}(\lambda_1) + \text{Poi}(\lambda_2) = \text{Poi}(\lambda_1 + \lambda_2)$.

Thinning (Splitting)

Each event type i w.p. p_i (indep.) $\Rightarrow$ type i: $\text{Poi}(\lambda p_i)$, **independent** of other types.

Conditioning Trick

Given sum of indep. Poissons $= n$, each count is Binomial:

$N_1 \mid (N_1 + N_2 = n) \sim \text{Bin}(n, \frac{\lambda_1}{\lambda_1 + \lambda_2})$

Split at deterministic time t, not random S_k.

Ex: $\mathbb{E}[S_4 \mid N(1) = 2] = 1 + 2/\lambda$ (2 more arrivals after $t = 1$).

NHPP

Intensity $\lambda(t)$. Mean value fn: $m(t) = \int_0^t \lambda(s)\, ds$.

$N(t) \sim \text{Poi}(m(t))$

$N(t) - N(s) \sim \text{Poi}(m(t) - m(s))$

Indep. increments: YES. Stationary increments: NO.

Interarrivals: **NOT independent, NOT identically distributed.**

$\mathbb{P}(T_2 > t \mid T_1 = s) = e^{-(m(s+t) - m(s))}$ depends on s.

NHPP Thinning: event at time s is type 1 w.p. $p(s)$ $\Rightarrow$ type 1: NHPP with intensity $\lambda(s)p(s)$.

NHPP Superposition: $\lambda_1(t) + \lambda_2(t)$; event at t from process 1 w.p. $\frac{\lambda_1(t)}{\lambda_1(t) + \lambda_2(t)}$.

Compound Poisson

$X(t) = \sum_{i=1}^{N(t)} Y_i, \quad Y_i$ iid, indep. of $N(t) \sim \text{Poi}(\lambda t)$.

$\mathbb{E}[X(t)] = \lambda t\, \mu_Y$

$\text{Var}(X(t)) = \lambda t\, \mathbb{E}[Y_i^2]$ (by Eve's Law)

CTMC — Setup

What It Means (Reframed)

A CTMC is built from two pieces:

1. How long in each state: In state i, stay for $\text{Exp}(v_i)$ time.

2. Where to go next: Upon leaving i, jump to j with prob $P_{ij} = q_{ij}/v_i$.

So: $q_{ij} = v_i P_{ij}$ = rate of $i \to j$ transitions; $v_i = \sum_{j \neq i} q_{ij}$.

Generator Matrix Q

$q_{ii} = -v_i, \quad q_{ij} \geq 0$ for $j \neq i$. **Rows sum to 0.**

Reading Q: given Q, read off $v_i = |q_{ii}|$ and $P_{ij} = q_{ij}/v_i$.

Transition Rates (Infinitesimal)

$q_{ij} = \lim_{h \to 0} P_{ij}(h)/h \quad (j \neq i); \quad q_{ii} = \lim_{h \to 0}(P_{ii}(h) - 1)/h = -v_i$.

Kolmogorov Equations

Forward: $\mathbf{P}'(t) = \mathbf{P}(t)\, Q$. Backward: $\mathbf{P}'(t) = Q\, \mathbf{P}(t)$.

Two-state solution $(q_{01} = \lambda, q_{10} = \mu)$: $P_{00}(t) = \frac{\mu + \lambda e^{-(\lambda+\mu)t}}{\lambda + \mu}, \quad P_{01}(t) = \frac{\lambda(1 - e^{-(\lambda+\mu)t})}{\lambda + \mu}$

Limiting/Stationary Probabilities

Global balance (always valid): $\boxed{v_j \pi_j = \sum_{k \neq j} q_{kj}\, \pi_k}$

Rate OUT of j = Rate INTO j. Plus $\sum \pi_j = 1$.

Time Reversibility

Detailed balance: $\pi_i q_{ij} = \pi_j q_{ji}$ for all i, j.

If detailed balance holds $\Rightarrow$ chain is time reversible.

Kolmogorov criterion: reversible iff for every cycle $i_1 \to \cdots \to i_k \to i_1$: $\prod q_{\text{clockwise}} = \prod q_{\text{counterclockwise}}$.

Reversed chain: $q^*_{ij} = \pi_j q_{ji}/\pi_i$. Same v_i.

All birth-death processes are time reversible.

Hitting Times (First-Step Analysis)

From state i to target set A: let $\tau_i = \mathbb{E}[\text{time to hit } A \mid X(0) = i]$.

$\tau_i = \frac{1}{v_i} + \sum_j P_{ij}\, \tau_j \quad (\tau_a = 0 \text{ for } a \in A)$

Birth-Death Process

CTMC where transitions: $i \to i+1$ (rate λ_i) or $i \to i-1$ (rate μ_i). $v_i = \lambda_i + \mu_i$.

Detailed balance: $\lambda_n P_n = \mu_{n+1} P_{n+1}$

$$\boxed{P_n = P_0 \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}}}$$

$P_0 = \left(1 + \sum_{n=1}^{\infty} \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}}\right)^{-1}$. Exists iff sum $< \infty$.

Duration Times $\mathbb{E}[T_i]$

T_i = time to go from i to 0. Recursive:
$\lambda_i \mathbb{E}[T_i] = 1 + \mu_i \mathbb{E}[T_{i-1}], \quad T_0 = 0.$

Queueing Models

System vs. Customer POV: L, W = system metrics; W = what a customer experiences. Little's Law connects them.

$$\boxed{L = \lambda_{\text{eff}} W, \quad L_q = \lambda_{\text{eff}} W_q}$$

PASTA: If arrivals are Poisson, arriving customers see the system in its **time-average** state: $a_n = P_n$.

M/M/1 ($\lambda < \mu$)

$\lambda_j = \lambda$, $\mu_j = \mu$. $\rho = \lambda/\mu$.
$P_n = (1-\rho)\rho^n \quad P_0 = 1 - \rho$
$L = \frac{\rho}{1-\rho} \quad W = \frac{1}{\mu-\lambda} \quad W_q = \frac{\rho}{\mu-\lambda} = \frac{\lambda}{\mu(\mu-\lambda)} \quad L_q = \frac{\rho^2}{1-\rho}$
Busy period: $\mathbb{E}[\text{Busy}] = \frac{1-P_0}{\lambda_0 P_0} = \frac{1}{\mu-\lambda}$

M/M/1/N (Finite Capacity N)

$P_n = \frac{1-\rho}{1-\rho^{N+1}} \rho^n, \quad n = 0, 1, \ldots, N$
(Works even if $\rho \geq 1$; no stability condition needed.)
Fraction lost $= P_N$ (PASTA). $\lambda_{\text{eff}} = \lambda(1 - P_N)$.

M/M/s (s servers, $\rho = \lambda/(s\mu) < 1$)

$\mu_j = \min(j, s)\mu$.

$$P_j = \begin{cases} \frac{(s\rho)^j}{j!} P_0 & j \leq s \\ \frac{s^s \rho^j}{s!} P_0 & j > s \end{cases}$$

$P_0 = \left[\sum_{k=0}^{s-1} \frac{(s\rho)^k}{k!} + \frac{(s\rho)^s}{s!(1-\rho)}\right]^{-1}$
$L_q = \frac{(s\rho)^s \rho}{s!(1-\rho)^2} P_0$

M/M/∞ (Infinite Servers)

$\lambda_j = \lambda$, $\mu_j = j\mu$. $P_n = e^{-\lambda/\mu} \frac{(\lambda/\mu)^n}{n!}$ (Poisson!).
$L = \lambda/\mu, \quad W_q = 0, \quad W = 1/\mu.$

M/M/s/K (Finite Capacity K)

Same as M/M/s but truncated at K. Normalize $P_0, \ldots, P_K$. Fraction lost $= P_K$.

Modeling Checklist

Scenario	Model/Tool
Exponential life-times, who fails first	Competing exponentials
Arrivals, counts, random times	Poisson process
Rate changes with time	NHPP
On/off cycles, proportion	Alternating renewal
Long-run rate of cost/reward	Renewal reward thm
Wait at random time	Inspection paradox / F_e
Arrivals + servers + queue	M/M/s queueing
Arrivals, full $\Rightarrow$ lost	M/M/s/K + PASTA
Machines + repairmen	CTMC (birth-death if symmetric rates; expanded state space otherwise)
System POV: how many present	$L = \sum n P_n$
Customer POV: how long to wait	$W = L/\lambda_{\text{eff}}$ (Little)

Is it birth-death? Only if transitions are ± 1. If different machines have different rates, counting state alone may not be Markov — expand state space.

Renewal Theory

$X_1, X_2, \ldots$ iid, $\mathbb{E}[X_i] = \mu$, $S_n = X_1 + \cdots + X_n$.
$N(t) = \max\{n : S_n \leq t\}, \quad m(t) = \mathbb{E}[N(t)]$.

Elementary Renewal Theorem

$\frac{N(t)}{t} \to \frac{1}{\mu}$ a.s., $\quad \frac{m(t)}{t} \to \frac{1}{\mu}$

Wald's Equation

X_i iid, N a stopping time, $\mathbb{E}[N] < \infty$:

$$\boxed{\mathbb{E}\left[\sum_{i=1}^{N} X_i\right] = \mathbb{E}[N] \cdot \mathbb{E}[X_1]}$$

Also: $\mathbb{E}[S_{N(t)+1}] = \mu(m(t) + 1)$.
$\mathbb{E}[\sum X_i \mid N = n] \neq n\mathbb{E}[X]$ in general! Conditioning on $N = n$ changes the distribution of summands.

Renewal Reward Theorem

R_n = reward in cycle n. Then:

$$\boxed{\text{Long-run rate} = \frac{\mathbb{E}[R]}{\mathbb{E}[X]} = \frac{\mathbb{E}[\text{reward/cycle}]}{\mathbb{E}[\text{cycle length}]}}$$

Recipe: (1) Identify cycle. (2) $\mathbb{E}$[cycle]. (3) $\mathbb{E}$[reward in cycle]. (4) Divide.

Alternating Renewal

ON (dur. Z) $\to$ OFF (dur. Y) $\to$ repeat. Cycle $= Z + Y$.

$$\boxed{\mathbb{P}(\text{ON at large } t) = \frac{\mathbb{E}[Z]}{\mathbb{E}[Z] + \mathbb{E}[Y]}}$$

Inspection Paradox & Equilibrium Dist.

At random time t (large), the interval containing t is **longer** than a typical X_i.

Equilibrium CDF: $F_e(y) = \frac{1}{\mu} \int_0^y \bar{F}(x)\, dx, \quad f_e(y) = \frac{\bar{F}(y)}{\mu}$.

Expected total observed interval: $\mathbb{E}[X_{N(t)+1}] \to \frac{\mathbb{E}[X^2]}{\mathbb{E}[X]} = \frac{\mathbb{E}[X^2]}{\mu}$

Expected excess (remaining) life: $\mathbb{E}[\gamma] = \frac{\mathbb{E}[X^2]}{2\mu}$ (also $= \mathbb{E}[\text{age}]$)

Always $\mathbb{E}[X^2]/\mu \geq \mu$ (Jensen). Equality iff X is constant.

Special cases:

- $X \sim \text{Exp}(\lambda)$: $F_e = F$. No paradox (memoryless).
- $X = c$ (constant): $F_e = \text{Unif}(0, c)$.

Key Connections to DTMC

$\pi = \pi P$ (DTMC)	$\leftrightarrow \pi Q = 0$ (CTMC)
$\mathbb{E}[T_{jj}] = 1/\pi_j$	same for both
First-step: $+1$ per step	$+1/v_i$ per holding
Period d (DTMC)	CTMC always aperiodic
Eve's Law	used for Var(W) everywhere

Eve's Law (Total Variance)

$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y|X)] + \text{Var}(\mathbb{E}[Y|X])$
Used in: compound Poisson Var($X(t)$), bus stop Var(W), Ehrenfest Var(X_n).